Task : Advanced Threat Hunting, SOAR Automation, and SOC Optimization

1. Objective

The objective of this week's training is to develop advanced Security Operations Center (SOC) capabilities, focusing on proactive threat hunting, automation using SOAR, post-incident analysis, adversary emulation, and security performance measurement.
This report covers both theoretical understanding and practical implementation of these concepts using industry-standard tools and simulated environments.
The practical section aims to apply advanced SOC techniques such as threat hunting methodologies, automation workflows, incident analysis, and adversary simulation to improve detection, response, and overall security posture.

2. Threat Hunting Methodologies

2.1 Overview

Threat hunting is a proactive cybersecurity approach focused on identifying hidden threats that evade traditional detection systems. Unlike reactive incident response, threat hunting is driven by hypotheses, attacker behavior analysis, and continuous monitoring of system activities.

2.2 Proactive Threat Hunting

Proactive hunting involves forming hypotheses based on attacker behavior and validating them using system and network data. For example, hunting for misuse of valid accounts (MITRE T1078) by analyzing unusual login patterns or privilege escalation attempts.
Key approaches include:

- Hypothesis-driven hunting (based on threat intelligence)
- Data-driven hunting (analyzing anomalies in logs)
- Intelligence-driven hunting (using external threat feeds)

2.3 Threat Hunting Frameworks

Structured methodologies improve hunting efficiency:

- **SqRR (Search, Query, Retrieve, Respond):**
  Focuses on iterative querying and refining results to identify threats.
- **TaHiTI (Targeted Hunting integrating Threat Intelligence):**
  Combines threat intelligence with internal telemetry for targeted investigations.
- **MITRE ATT&CK-based Hunting:**
  Uses known adversary TTPs (Tactics, Techniques, Procedures) to guide hunting activities.

2.4 Data Sources for Threat Hunting

Effective threat hunting relies on multiple data sources:

- Endpoint logs (EDR data)
- Network traffic (NetFlow, DNS logs)
- Authentication logs (Event ID 4624, 4625)
- SIEM platforms like Elastic Security
- Threat intelligence feeds (e.g., AlienVault OTX)

2.5 Advanced Techniques

- Behavioral analytics for detecting insider threats
- Baseline comparison to identify anomalies
- Detection of Living-off-the-Land (LotL) attacks
- Use of threat hunting queries (KQL, SPL)

2.6 Key Objective

To develop the ability to proactively identify advanced threats using structured methodologies, threat intelligence, and deep log analysis.

3. Advanced SOAR Automation

3.1 Overview

SOAR (Security Orchestration, Automation, and Response) enhances SOC efficiency by automating repetitive tasks and orchestrating workflows across multiple security tools.

3.2 SOAR Components

- **Orchestration:** Integration of tools like SIEM, EDR, and threat intelligence platforms
- **Automation:** Execution of predefined actions (e.g., auto-block IP, auto-ticket creation)
- **Response:** Automated or semi-automated incident handling

3.3 Playbook Development

Playbooks define step-by-step automated workflows for incident handling.

Examples:

- Phishing response playbook (email analysis → link scan → user notification)
- Malware response playbook (file hash → VirusTotal check → isolate endpoint)

Tools like TheHive support structured playbook execution.

3.4 Integration with Security Tools

SOAR integrates with:

- SIEM tools like Wazuh
- Threat intelligence platforms
- Ticketing systems

This enables automated alert enrichment and faster response.

3.5 Advanced Automation Concepts

- Auto-enrichment of alerts using APIs
- Conditional workflows based on severity
- Automated containment (blocking IP, disabling accounts)
- Integration with SOAR tools like Splunk SOAR

3.6 Key Objective

To automate SOC workflows, reduce manual effort, and improve response speed and consistency.

4. Post-Incident Analysis and Continuous Improvement

4.1 Overview

Post-incident analysis focuses on identifying the root cause of incidents and improving security posture through structured review processes.

4.2 Root Cause Analysis (RCA)

RCA identifies the underlying reason behind an incident.
Common techniques:

- **5 Whys Method:** Iteratively asking “why” to reach root cause
- **Fishbone Diagram:** Categorizing causes (people, process, technology)

Example: Phishing attack → user clicked link → lack of awareness → inadequate training

4.3 Lessons Learned Process

After incident resolution:

- Review what worked and what failed
- Identify gaps in detection and response
- Update playbooks and controls

4.4 SOC Metrics and KPIs

Key performance indicators include:

- Mean Time to Detect (MTTD)
- Mean Time to Respond (MTTR)
- False positive rate
- Incident resolution time

These metrics help evaluate SOC efficiency.

4.5 Advanced Improvements

- Updating SIEM rules based on incidents
- Enhancing detection signatures
- Conducting regular tabletop exercises
- Continuous analyst training

4.6 Key Objective

To improve SOC operations by analyzing incidents, identifying weaknesses, and implementing continuous improvements.

5. Adversary Emulation Techniques

5.1 Overview

Adversary emulation involves simulating real attacker behavior to test detection and response capabilities of a SOC.

5.2 Emulation of Attacker Techniques

Examples:

- Phishing attacks (MITRE T1566)
- Remote service exploitation (T1210)
- Credential dumping (T1003)

These simulations help identify detection gaps.

5.3 Emulation Frameworks

- An MITRE Caldera
- Atomic Red Team
- Metasploit Framework

These tools simulate real-world attack scenarios.

5.4 Red Team vs Blue Team

- **Red Team:** Simulates attackers
- **Blue Team:** Defends and detects
- **Purple Team:** Collaboration between both

This improves overall security posture.

5.5 Advanced Concepts

- Continuous adversary simulation
- Detection validation
- Rule tuning based on emulation results

5.6 Key Objective

To simulate attacker behavior and validate SOC detection and response capabilities.

6. Security Metrics and Executive Reporting

6.1 Overview

Security metrics measure SOC performance and help communicate security posture to management and stakeholders.

6.2 Advanced SOC Metrics

- Dwell Time (time attacker remains undetected)
- Detection rate
- False positive rate
- Incident closure rate

6.3 Executive Reporting

Reports should:

- Be clear and non-technical
- Use graphs and dashboards
- Highlight risks and business impact

6.4 Visualization and Communication

Tools like dashboards help present:

- Incident trends
- Severity distribution
- Response timelines

6.5 Continuous Improvement Using Metrics

Metrics help:

- Identify weak areas
- Improve detection speed
- Optimize SOC processes

6.6 Advanced Considerations

- Risk-based reporting
- Mapping incidents to business impact
- Compliance reporting (ISO 27001, NIST)

6.7 Key Objective

To measure SOC performance effectively and communicate insights to both technical and non-technical stakeholders.

7. Threat Hunting Practice

7.1 Objective

The objective of this task is to perform proactive threat hunting by developing a hypothesis, analyzing logs, and validating findings using threat intelligence sources.

7.2 Methodology

A hypothesis-driven threat hunting approach was used to identify potential misuse of valid accounts. Logs were analyzed and correlated with threat intelligence sources to validate suspicious activity.

7.3 Hypothesis

A hypothesis was developed stating that unauthorized privilege escalation may occur through misuse of valid accounts (MITRE T1078).

Hypothesis: There may be unauthorized privilege escalation activity on the system through misuse of valid accounts (T1078).

Fig 1: Threat Hunting Hypothesis

7.4 Log Analysis

System logs were analyzed to identify privileged access events (Event ID 4672). Suspicious entries were identified where unexpected users were assigned administrative privileges.

Threat Hunting Log Analysis

Timestamp	User	Event ID	Notes
2025-04-08 15:00:00	testuser	4672	Unexpected admin role
2025-04-08 15:05:00	admin1	4672	Normal activity

Fig 2: Privilege Escalation Log Analysis

7.5 Threat Intelligence Validation

Threat intelligence was gathered using AlienVault OTX to identify indicators related to valid account misuse. Relevant patterns and indicators were reviewed.

Fig 3: Threat Intelligence Lookup in AlienVault OTX

7.6 Endpoint Analysis

Velociraptor was used to analyze running processes and identify any abnormal or suspicious activity on the endpoint.

Fig 4: Process Analysis using Velociraptor

7.7 Correlation and Findings

Log data, threat intelligence, and endpoint activity were correlated to identify potential suspicious behavior. The findings suggest possible misuse of valid accounts.

7.8 Hunting Report

Threat hunting was conducted based on the hypothesis of unauthorized privilege escalation using valid accounts (MITRE T1078). Log analysis identified Event ID 4672 entries indicating elevated privileges. Threat intelligence from AlienVault OTX provided context on potential misuse patterns. Endpoint analysis using Velociraptor showed active processes during the event timeframe. Correlation of all data sources suggested possible

suspicious activity involving privilege escalation. Although no confirmed compromise was detected, the findings highlight the importance of monitoring privileged access and validating user activity to prevent unauthorized access.

7.9 Key Objective

The key objective of this task is to develop the ability to perform proactive threat hunting using hypothesis-driven methodologies, correlate multiple data sources, and validate findings using threat intelligence to identify potential security threats effectively.

8. SOAR Playbook Development

8.1 Overview

This task is to design and simulate a SOAR playbook for automated incident response to improve efficiency and reduce response time.

8.2 Playbook Design

A phishing response playbook was created to automate detection, validation, and response actions. The playbook includes steps for IP reputation checking, blocking malicious IPs, and creating incident tickets.

Playbook Name: Phishing Alert Response

Step 1: Receive alert from SIEM (Wazuh)
Step 2: Extract IP address from alert
Step 3: Check IP reputation using VirusTotal / OTX
Step 4: If malicious → Block IP using CrowdSec
Step 5: Create incident ticket in TheHive
Step 6: Assign to SOC Analyst (Tier 2 if critical)
Step 7: Document response and close incident

Fig 5: SOAR Playbook Design

8.3 Alert Simulation

A mock phishing alert was generated to simulate a real-world scenario involving a suspicious IP address and file hash.

```
Alert: Phishing Email Detected
Source IP: 192.168.40.140
File Hash: abcd1234efgh5678
Severity: High
```

Fig 6: Simulated Phishing Alert

8.4 Threat Intelligence Validation

Based on the validation results, the alert was confirmed as suspicious and required further investigation. This demonstrates how threat intelligence supports accurate decision-making.

Fig 7: IP Reputation Check

8.5 Response Execution

Based on validation results, the IP was blocked (simulated) and appropriate response actions were taken as per the playbook.

```
┌──(kali㉿kali)-[~]
└─$ ping 192.168.40.140
PING 192.168.40.140 (192.168.40.140) 56(84) bytes of data.
From 192.168.40.139 icmp_seq=1 Destination Host Unreachable
From 192.168.40.139 icmp_seq=2 Destination Host Unreachable
From 192.168.40.139 icmp_seq=3 Destination Host Unreachable
From 192.168.40.139 icmp_seq=4 Destination Host Unreachable
ping: sendmsg: No route to host
From 192.168.40.139 icmp_seq=5 Destination Host Unreachable
From 192.168.40.139 icmp_seq=6 Destination Host Unreachable
^C
--- 192.168.40.140 ping statistics ---
10 packets transmitted, 0 received, +6 errors, 100% packet loss, time 9217ms
pipe 3
```

Fig 8: Simulated IP Blocking

8.6 Incident Documentation

An incident ticket was created to document the alert, response actions, and assignment details.

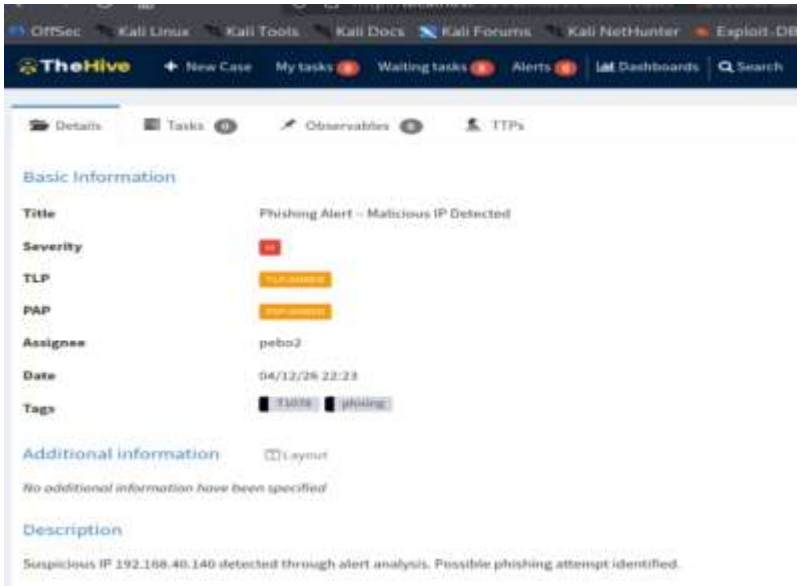

Fig 9: Incident Ticket

8.7 Playbook Execution Results

The execution of the playbook was documented, showing successful completion of all steps.

Playbook Step	Status	Notes
Check IP	Success	IP checked in OTX
Block IP	Success	Simulated CrowdSec blocking
Create Ticket	Success	Incident documented

Fig 10: Playbook Execution Results

8.8 Playbook Summary

A SOAR playbook was developed to automate phishing incident response. The playbook includes IP reputation checking, automated blocking, and ticket creation. Simulation demonstrated improved response efficiency and consistency. This approach reduces manual effort and ensures standardized handling of security incidents within the SOC environment.

8.9 Key Objective

The key objective of this task is to design and implement automated response workflows using SOAR concepts to enhance SOC efficiency and ensure consistent incident handling.

9. Post-Incident Analysis

9.1 Overview

This task is to analyze a security incident, identify its root cause, and evaluate SOC performance using key metrics.

9.2 Incident Overview

A phishing email attack was simulated where a user clicked a malicious link, leading to unauthorized access.

Incident: Phishing Email Attack
User received a phishing email and clicked a malicious link, leading to unauthorized access.

Fig 11: Incident Scenario Description

9.3 Root Cause Analysis

The 5 Whys technique was used to identify the root cause of the incident. The analysis revealed that lack of user awareness and weak security policies were the primary causes.

Question	Answer
Why was email opened?	User clicked malicious email
Why was link clicked?	Lack of awareness
Why lack of awareness?	No security training
Why no training?	Weak security policies
Why weak policies?	Lack of security governance

Fig 12: Root Cause Analysis using 5 Whys

9.4 Fishbone Diagram Analysis

A Fishbone Diagram was created to categorize the causes into people, process, and technology factors. This helped in understanding the broader impact and contributing factors.

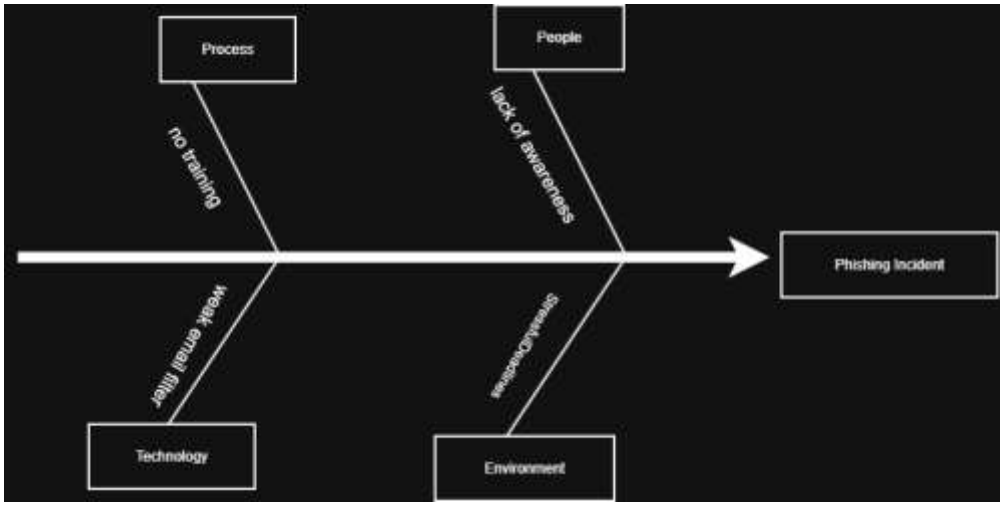

Fig 13: Fishbone Diagram for Incident Analysis

9.5 SOC Metrics Evaluation

SOC performance metrics were calculated based on the incident timeline. MTTD was 2 hours and MTTR was 4 hours.

Detection Time: 2 hours
Response Time: 4 hours

MTTD (Mean Time to Detect) = 2 hours
MTTR (Mean Time to Respond) = 4 hours

Fig 14: SOC Metrics Calculation

9.6 Analysis and Findings

The incident analysis showed gaps in user awareness and response efficiency. Improvements are required in training and detection mechanisms.

The detection time indicates moderate monitoring capability, while response time suggests improvement is needed in response processes.

Fig 15: Metrics Interpretation

9.7 Metrics Summary

The incident analysis revealed that detection occurred within 2 hours, while response took 4 hours. These metrics indicate moderate SOC performance. Improvements in monitoring and faster response mechanisms are required. Regular training and improved detection tools can help reduce response time and enhance overall security operations efficiency.

9.8 Key Objective

The key objective of this task is to perform structured incident analysis, identify root causes, and use metrics to improve SOC performance and response strategies.

10. Alert Triage with Automation

10.1 Overview

This task is to perform alert triage and automate validation using threat intelligence to improve detection accuracy and response efficiency.

10.2 Alert Simulation

A simulated alert was created representing a suspicious file download from an internal system. The alert included details such as IP address, file hash, and severity level.

Alert ID: 005
Description: Suspicious File Download Detected
Source IP: 192.168.40.140
Destination: Internal Server
File Hash: e3b0c44298fc1c149afbf4c8996fb924
Severity: High
Status: Open

Fig 16: Simulated Security Alert

10.3 Triage Analysis

The alert was analyzed to determine its severity and potential impact. Based on the suspicious activity and unknown file hash, the alert was classified as high priority.

Initial Analysis:
- Suspicious file download detected
- Unknown file hash
- Internal system targeted

Decision:
- Priority set to High
- Requires immediate validation

Fig 17: Alert Triage Analysis

10.4 Threat Intelligence Validation

The file hash was validated using VirusTotal to determine whether it is malicious or benign. This helped in making an informed decision regarding the alert.

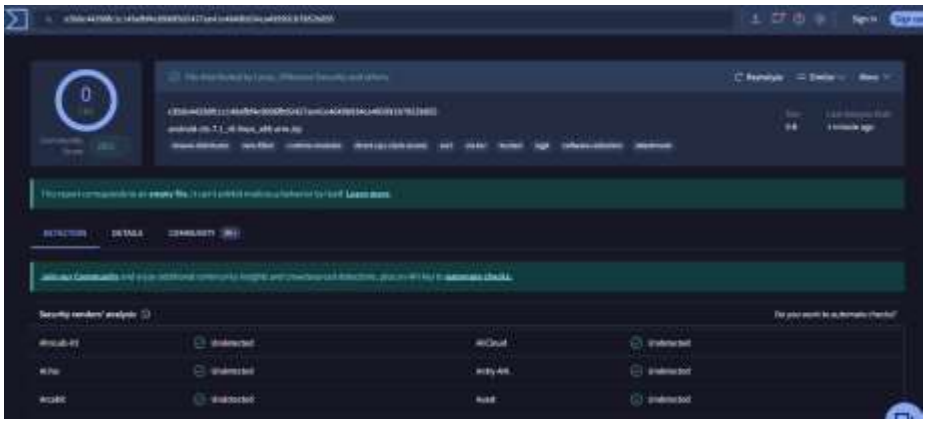

Fig 18: File Hash Validation in VirusTotal

10.5 Automation Workflow

An automated workflow was simulated where the alert is automatically enriched using threat intelligence and forwarded for incident handling.

Automation Workflow:
- Alert received from SIEM
- Hash automatically sent to VirusTotal
- Result analyzed
- Ticket created in TheHive
- Alert assigned to SOC Analyst

Fig 19: Automated Validation Workflow

10.6 Incident Documentation

An incident ticket was created to document the alert and assign it for further investigation.

Title: Suspicious File Download
IP: 192.168.40.140
Hash: e3b0c44298fc1c149afbf4c8996fb924
Priority: High
Status: Open
Assigned: SOC Analyst

Fig 20: Incident Ticket Creation

10.7 Triage Results

The final triage details were documented, confirming the alert priority and current investigation status.

Alert ID	Description	Source IP	Priority	Status
005	File Download	192.168.40.140	High	Open

Fig 21: Alert Triage Table

10.8 Summary

The alert triage process identified a suspicious file download and validated it using VirusTotal. Automation was simulated to enrich the alert and create an incident ticket. This approach improves efficiency, reduces manual effort, and ensures consistent analysis of security alerts in a SOC environment.

10.9 Key Objective

The key objective of this task is to perform efficient alert triage and leverage automation for threat intelligence validation to enhance SOC operations.

11. Evidence Analysis

11.1 Overview

This task is to collect, analyze, and preserve digital evidence while maintaining integrity and proper documentation.

11.2 Evidence Collection

Network data was collected using Velociraptor to identify active connections and system activity.

Fig 22: Velociraptor Dashboard

Fig 23: Network Connections Collection

11.3 Evidence Analysis

The collected data was analyzed to identify suspicious connections. External IP communication was observed, indicating potential unauthorized activity.

1124	velocirapt	IPv4	TCP	ESTAB	127.0.0.1	8889	127.0.0.1	51854	2026-04-12T18:38:20Z
664	lsass.exe	IPv4	TCP	LISTEN	0.0.0.0	49664	0.0.0.0	0	2026-04-12T14:58:27Z
500	wininit.exe	IPv4	TCP	LISTEN	0.0.0.0	49665	0.0.0.0	0	2026-04-12T14:58:27Z
1384	svchost.ex	IPv4	TCP	LISTEN	0.0.0.0	49666	0.0.0.0	0	2026-04-12T14:58:29Z
1400	svchost.ex	IPv4	TCP	LISTEN	0.0.0.0	49667	0.0.0.0	0	2026-04-12T14:58:29Z
2872	spoolsv.ex	IPv4	TCP	LISTEN	0.0.0.0	49668	0.0.0.0	0	2026-04-12T14:58:34Z
644	services.e	IPv4	TCP	LISTEN	0.0.0.0	49669	0.0.0.0	0	2026-04-12T14:58:39Z
1124	velocirapt	IPv4	TCP	ESTAB	127.0.0.1	50140	127.0.0.1	8000	2026-04-12T15:09:48Z
1124	velocirapt	IPv4	TCP	ESTAB	127.0.0.1	50144	127.0.0.1	8000	2026-04-12T15:09:48Z
1124	velocirapt	IPv4	TCP	ESTAB	127.0.0.1	50146	127.0.0.1	8000	2026-04-12T15:09:48Z
1124	velocirapt	IPv4	TCP	ESTAB	127.0.0.1	50148	127.0.0.1	8000	2026-04-12T15:09:49Z
3896	svchost.ex	IPv4	TCP	ESTAB	192.168.4	51665	4.213.25.241	443	2026-04-12T18:27:25Z
1124	velocirapt	IPv4	TCP	ESTAB	127.0.0.1	51678	127.0.0.1	8001	2026-04-12T18:31:24Z
6300	SearchApp	IPv4	TCP	CLOSE_W	192.168.4	51732	23.15.33.55	443	2026-04-12T18:33:10Z
6300	SearchApp	IPv4	TCP	CLOSE_W	192.168.4	51770	96.17.104.232	443	2026-04-12T18:34:22Z
0		IPv4	TCP	TIME_WA	192.168.4	51846	72.145.35.100	443	1601-01-01T00:00:00Z
0		IPv4	TCP	TIME_WA	192.168.4	51847	23.50.14.43	80	1601-01-01T00:00:00Z
0		IPv4	TCP	TIME_WA	192.168.4	51850	23.50.14.43	80	1601-01-01T00:00:00Z
4340	msedge.ex	IPv4	TCP	ESTAB	127.0.0.1	51854	127.0.0.1	8889	2026-04-12T18:38:20Z
0		IPv4	TCP	TIME_WA	192.168.4	51855	72.145.35.118	443	1601-01-01T00:00:00Z
0		IPv4	TCP	TIME_WA	192.168.4	51856	72.145.35.104	443	1601-01-01T00:00:00Z

Fig 24: Suspicious Network Activity Identified

11.4 Process Analysis

System processes were analyzed to identify any abnormal or suspicious processes running on the system.

Fig 25: Process Analysis using Velociraptor

11.5 Evidence Integrity

A SHA256 hash was generated for collected evidence to ensure integrity and prevent tampering.

```
C:\Users\hardi>certutil -hashfile sample.txt SHA256
SHA256 hash of sample.txt:
6722aeed5f5d8297de4eb2cba1c5abb900afe4a4cd64b10e2835463087936666
CertUtil: -hashfile command completed successfully.
```

Fig 26: SHA256 Hash Generation

11.6 Evidence Preservation

FTK Imager was used to securely handle and document evidence for forensic analysis.

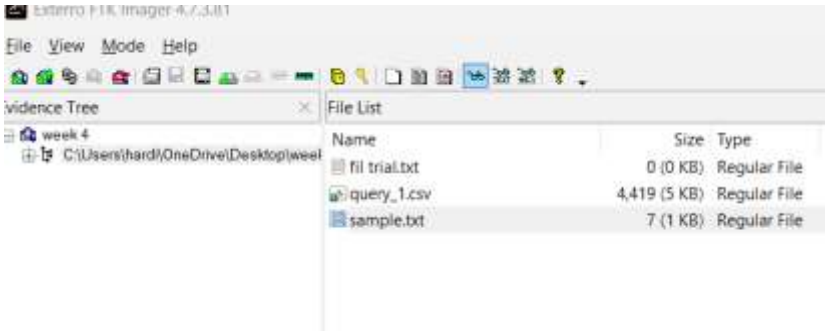

Fig 27: Evidence Added in FTK Imager

11.7 Chain of Custody

A chain-of-custody record was maintained to track evidence handling and ensure accountability.

Item	Description	Collected By	Date	Hash Value
Network Log	query1_csv	SOC Analyst	2025-08-18	N/A
File	sample.txt	SOC Analyst	2025-08-18	6722aeed5f5d8297de4eb2cba1c5abb900afe4a4cd64b10e283546308793666 6

Fig 28: Chain of Custody Documentation

11.8 Summary

Evidence analysis was performed using Velociraptor to collect network data and identify suspicious activity. Evidence integrity was ensured through SHA256 hashing. FTK Imager was used for documentation and handling. Proper chain-of-custody was maintained, ensuring secure and reliable forensic investigation within the SOC environment.

11.9 Key Objective

The key objective of this task is to perform evidence collection, analysis, and preservation while maintaining integrity and proper documentation for forensic investigations.

12. Adversary Emulation Practice

12.1 Overview

This task is to simulate adversary techniques and evaluate SOC detection capabilities.

12.2 Environment Setup

Kali Linux was used as the attacker machine and Metasploitable2 as the target system.

Fig 29: Attacker and Target IP Configuration

12.3 Attack Simulation

Basic attack behavior was simulated using network scanning and communication techniques to mimic phishing-related activity.

Fig 30: Simulated Attack Activity from Kali

12.4 Target System Activity

Network activity on the target system was analyzed to observe incoming connections from the attacker.

```
tcp        0      0 0.0.0.0:5900            0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:111             0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:6000            0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:80              0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:32817           0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:36882           0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:8787            0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:8180            0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:1524            0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:21              0.0.0.0:*               LISTEN
tcp        0      0 192.168.40.140:53       0.0.0.0:*               LISTEN
tcp        0      0 127.0.0.1:53            0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:23              0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:5432            0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:25              0.0.0.0:*               LISTEN
tcp        0      0 127.0.0.1:953           0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:445             0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:37693           0.0.0.0:*               LISTEN
tcp6       0      0 :::2121                 :::*                    LISTEN
tcp6       0      0 :::3632                 :::*                    LISTEN
tcp6       0      0 :::53                   :::*                    LISTEN
tcp6       0      0 :::22                   :::*                    LISTEN
```

Fig 31: Network Activity on Target System

12.5 Detection Simulation

SOC performance metrics were calculated based on the incident timeline. MTTD was 2 hours and MTTR was 4 hours.

Alert: Suspicious External Connection Detected
Source IP: 192.168.40.139
Target IP: 192.168.40.140
Technique: T1566 (Phishing Simulation)
Status: Detected

Fig 32: Simulated Detection Alert

12.6 Detection Results

The detection results were documented, showing successful identification of suspicious activity.

Timestamp	TTP	Detection Status	Notes
2025-04-10 17:00:00	T1566	Detected	Suspicious activity observed

Fig 33: Detection Results Table

12.7 Analysis

The simulation demonstrated how adversary techniques can be detected through monitoring network behavior and system activity.

The simulated attack activity demonstrated how an attacker can initiate communication with a target system. Detection mechanisms identified unusual network behavior, indicating possible malicious intent.

Fig 34: Emulation Analysis

12.8 Emulation Report

Adversary emulation was performed by simulating phishing-related attack behavior using network scanning and communication techniques. Kali Linux was used to initiate activity against the Metasploitable2 target system. The generated traffic was analyzed, and detection was simulated to represent SOC monitoring capabilities. The activity was mapped to MITRE ATT&CK technique T1566. The results showed that unusual network behavior can be identified through proper monitoring and analysis. This exercise highlights the importance of proactive detection mechanisms and continuous monitoring in identifying potential threats and improving SOC readiness against real-world attack scenarios.

12.9 Key Objective

The key objective of this task is to simulate adversary techniques and evaluate detection capabilities to strengthen SOC defense mechanisms.

13. Security Metrics and Executive Reporting

13.1 Overview

This task is to calculate SOC performance metrics and present findings in a structured executive report.

13.2 Data Collection

Incident data including detection, response, and resolution times was collected and organized for analysis.

Incident ID	Detection Time	Response Time	Resolution Time
001	10:00 AM	12:00 PM	02:00 PM
002	11:00 AM	01:00 PM	03:00 PM

Fig 35: Incident Data for Metrics Calculation

13.3 Metrics Calculation

SOC metrics such as MTTD, MTTR, and dwell time were calculated based on the collected data.

MTTD (Mean Time to Detect): 2 hours
MTTR (Mean Time to Respond): 2 hours
Dwell Time: 4 hours

Fig 36: SOC Metrics Calculation

13.4 Dashboard Visualization

A dashboard was created to visually represent SOC performance metrics for better understanding.

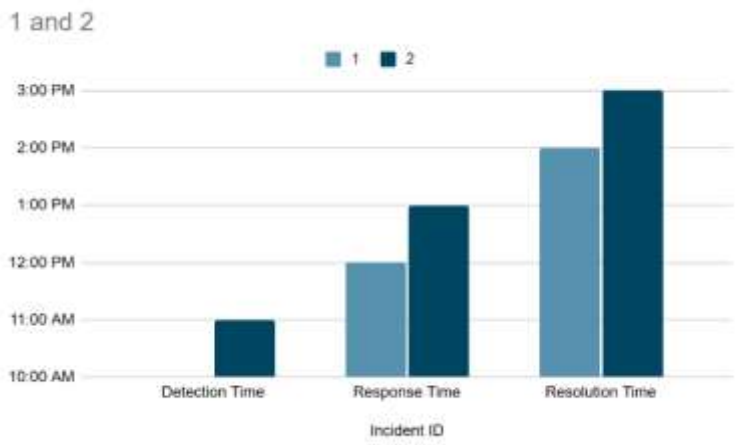

Fig 37: Metrics Dashboard Visualization

13.5 Metrics Analysis

The calculated metrics were analyzed to identify strengths and areas for improvement in SOC operations.

METRICS ANALYSIS :
The calculated SOC metrics indicate moderate detection and response efficiency. MTTD and MTTR values suggest timely identification and handling of incidents. However, dwell time highlights potential delays in complete resolution. Improving monitoring systems and automation can further reduce response time and enhance overall SOC performance and effectiveness.

Fig 38: Metrics Analysis Summary

13.6 Executive Reporting

An executive summary was prepared to present findings in a clear and concise manner for non-technical stakeholders.

Executive Summary:

The Security Operations Center (SOC) performance was evaluated using key metrics including Mean Time to Detect (MTTD), Mean Time to Respond (MTTR), and dwell time. The analysis revealed that incidents were detected and responded to within an average of two hours, indicating effective monitoring and response capabilities. However, the dwell time suggests that further improvements are required in incident resolution processes.

The SOC demonstrated strong capabilities in identifying and responding to threats, but opportunities exist to enhance efficiency through automation and improved threat intelligence integration. Implementing advanced detection mechanisms and optimizing response workflows can significantly reduce incident handling time.

It is recommended to invest in automation tools and continuous monitoring strategies to improve SOC performance. Regular evaluation of metrics and process improvements will ensure a proactive and resilient security posture.

Fig 39: Executive Report in Google Docs

13.7 Summary

SOC metrics were calculated and analyzed to evaluate performance. A dashboard was created for visualization, and an executive report was prepared. The results highlighted efficient detection and response while identifying areas for improvement in resolution time. This enhances decision-making and strengthens overall security operations.

13.8 Key Objective

The key objective of this task is to measure SOC performance and effectively communicate findings to stakeholders through structured reporting.

14. Capstone Project: Comprehensive SOC Workflow

14.1 Overview

This task is to simulate a complete SOC workflow including attack detection, triage, response, analysis, and reporting.

14.2 Attack Simulation

A Samba exploitation attack was simulated using Metasploit against the Metasploitable2 target system.

14.3 Adversary Emulation

Adversary behavior was simulated using network scanning and communication techniques.

14.4 Detection and Triage

The attack was detected and classified as a critical alert. Triage was performed to prioritize the incident.

Alert: Samba Exploit Detected
Source IP: 192.168.40.139
Target IP: 192.168.40.140
Technique: T1210
Severity: Critical

Fig 40: Detection Alert

Alert ID	Description	Source IP	Priority	Status
006	Samba Exploit	192.168.1.101	Critical	Open

Fig 41: Alert Triage

14.5 Response and Containment

The affected system was isolated and attacker access was blocked to prevent further damage.

Response Action:
- Target system isolated
- Attacker IP blocked
- Access terminated

Fig 42: Response Action

14.6 Automation Workflow

A simulated SOAR workflow was used to automate alert handling and response actions.

Automation Workflow:
- Alert triggered
- IP reputation checked
- Ticket created
- Auto-block initiated

Fig 43: SOAR Workflow

14.7 Post-Incident Analysis

Root cause analysis identified system vulnerabilities and lack of patching as key issues.

Why attack happened? → Vulnerable Samba service
Why vulnerable? → Outdated system
Why outdated? → No patching
Why no patching? → Weak policy
Why weak policy? → Lack of monitoring

Fig 44: Root Cause Analysis

14.8 Metrics Evaluation

SOC metrics were calculated to evaluate performance and response efficiency.

MTTD: 2 hours
MTTR: 2 hours
Dwell Time: 4 hours

Fig 45: Metrics Dashboard

14.9 Final Report

Executive Summary

A simulated cyberattack was conducted targeting a vulnerable Samba service on a Metasploitable2 system. The attack was initiated using Metasploit from a Kali Linux machine, representing a real-world exploitation scenario. The Security Operations Center (SOC) successfully detected the activity, classified it as critical, and initiated response procedures. Immediate containment actions were taken to isolate the affected system and block the attacker's access. The incident highlights the importance of proactive monitoring, timely patching, and automated response mechanisms.

Timeline

- **15:00 –** Attack initiated using Metasploit (Samba exploit)
- **15:05 –** Suspicious activity detected (simulated alert generated)
- **15:10 –** Alert triaged and classified as Critical
- **15:15 –** Response initiated (system isolation and IP blocking)
- **15:30 –** Investigation and analysis completed

Root Cause Analysis (RCA)

The root cause of the incident was an unpatched Samba vulnerability present on the target system. The system lacked regular updates and patch management, making it susceptible to exploitation. Additionally, weak monitoring policies delayed proactive detection. The absence of automated patching and vulnerability management contributed significantly to the successful execution of the attack.

Recommendations

- Implement regular patch management and vulnerability scanning

- Deploy continuous monitoring and alerting mechanisms
- Integrate automated response solutions (SOAR)
- Strengthen access controls and network segmentation
- Conduct regular security assessments and training

14.10 Stakeholder Briefing

A simulated cyberattack targeting a vulnerable system was successfully detected and managed by the SOC team. The attack exploited a known vulnerability in the system, allowing unauthorized access. The security team quickly identified the threat and classified it as critical. Immediate actions were taken to isolate the affected system and block the attacker's access.

The investigation revealed that the vulnerability existed due to outdated software and lack of timely patching. The response process was efficient, but improvements can be made in reducing response time and strengthening preventive measures. Automation can further enhance response capabilities and reduce manual effort.

It is recommended to implement regular system updates, improve monitoring systems, and adopt automated security solutions to prevent similar incidents in the future. Overall, the SOC demonstrated effective detection and response capabilities.

14.11 Key Objective

The key objective of this task is to simulate and understand a complete SOC workflow from attack detection to final reporting and improvement.

15. Challenges Faced

- Managing system resources efficiently while running multiple tools and virtual machines simultaneously
- Initial learning curve in understanding and using tools like Velociraptor for evidence collection
- Troubleshooting installation and execution issues for security tools and virtual environme

16. Key Learnings

- Gained practical experience in using SIEM tools such as Elastic Security and Wazuh for log analysis and correlation
- Understood incident escalation processes and case handling using TheHive
- Practiced evidence collection, analysis, and integrity verification using Velociraptor and hashing techniques
- Developed the ability to execute a complete SOC workflow including detection, triage, response, and reporting

17. Conclusion

This report demonstrates a comprehensive understanding of advanced SOC operations through both theoretical concepts and practical implementation. Key areas such as threat hunting, incident response, escalation workflows, and security monitoring were successfully explored.

The use of real tools along with simulated environments provided valuable hands-on experience and a clear understanding of real-world SOC processes. Overall, this training significantly enhanced technical knowledge, analytical skills, and preparedness for cybersecurity roles in a professional SOC environment.